MILITARY SPECIFICATION

RADIACMETER, IM-239/WDQ

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the design and test requirements for the IM-239/WDQ Radiacmeter, Air Monitor for Radioactive Particles, hereinafter referred to as the radiacmeter.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue the listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-15303	Audible Signals: Alarms, Bells, Buzzers, Horns, Sirens, And Electronic Shipboard
MIL-E-16400	Electronic, Interior Communication And Navigation Equipment, Naval Ship And Shore: General Specification For
MIL-E-17555	Electronic And Electrical Equipment, Accessories, And Repair Parts; Packaging And Packing Of
MIL-R-28760	Radiacmeter, Air Particle Detector, IM-192/WDQ

STANDARDS

MILITARY

MIL-STD-12	Abbreviations For Use On Drawings, And In Specifications, Standards And Technical Documents
MIL-STD-108	Definition Of And Basic Requirements For Enclosure For Electric And Electronic Equipment
MIL-STD-109	Quality Assurance Terms And Definitions
MIL-STD-415	Test Provisions For Electronic Systems And Associated Equipment, Design Criteria For
MIL-STD-461	Electromagnetic Emission And Susceptibility Requirements For The Control Of Electromagnetic Interference
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement Of
MIL-STD-781	Reliability Design Qualification And Production Acceptance Tests: Exponential Distribution
MIL-STD-810	Environmental Test Methods And Engineering Guidelines
MIL-STD-965	Parts Control Program
MIL-STD-1364	Standard General Purpose Electronic Test Equipment
DoD-STD-1399, Section 300	Interface Standard For Shipboard Systems, Electric Power, Alternating Current (Metric)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command (ELEX-8111), Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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MIL-STD-1472	Human Engineering Design Criteria For Military Systems, Equipment And Facilities
DoD-STD-1679	Software Development
MS16142	Boss, Gasket Seal Straight Thread, Tube Fitting; Standard Dimensions For

HANDBOOKS

MILITARY

MIL-HDBK-241	Design Guide For EMI Reduction In Power Supplies

(Copies of specifications, standards and handbooks, required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. The issue of the document which is indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA-70-1983	National Electrical Code

(Application for copies should be addressed to the National Fire Protection Association, 470 Atlantic Avenue, Boston, MA 02210.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General. The radiacmeter shall be in accordance with MIL-E-16400 to the extent specified herein.

3.1.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.2 Radiacmeter description. The radiacmeter is a detection and monitoring system for radioactive particles suspended in the air. The radiacmeter is to be used whenever a continuous and accurate measurement of radioactive air particulate is required to ensure that these radiation levels remain within safe levels. Airborne particles of a certain size containing radiation-emitting alpha and beta particles are of the greatest concern. The radiacmeter shall be designed to have its greatest sensitivity to beta radiation, as specified herein, while discriminating against a gamma background. Any sensitivity to alpha particles is desirable and shall be retained in the design whenever possible. Radioactive gas measurement shall not be a primary function of the radiacmeter because such measurements are to be provided by other equipment.

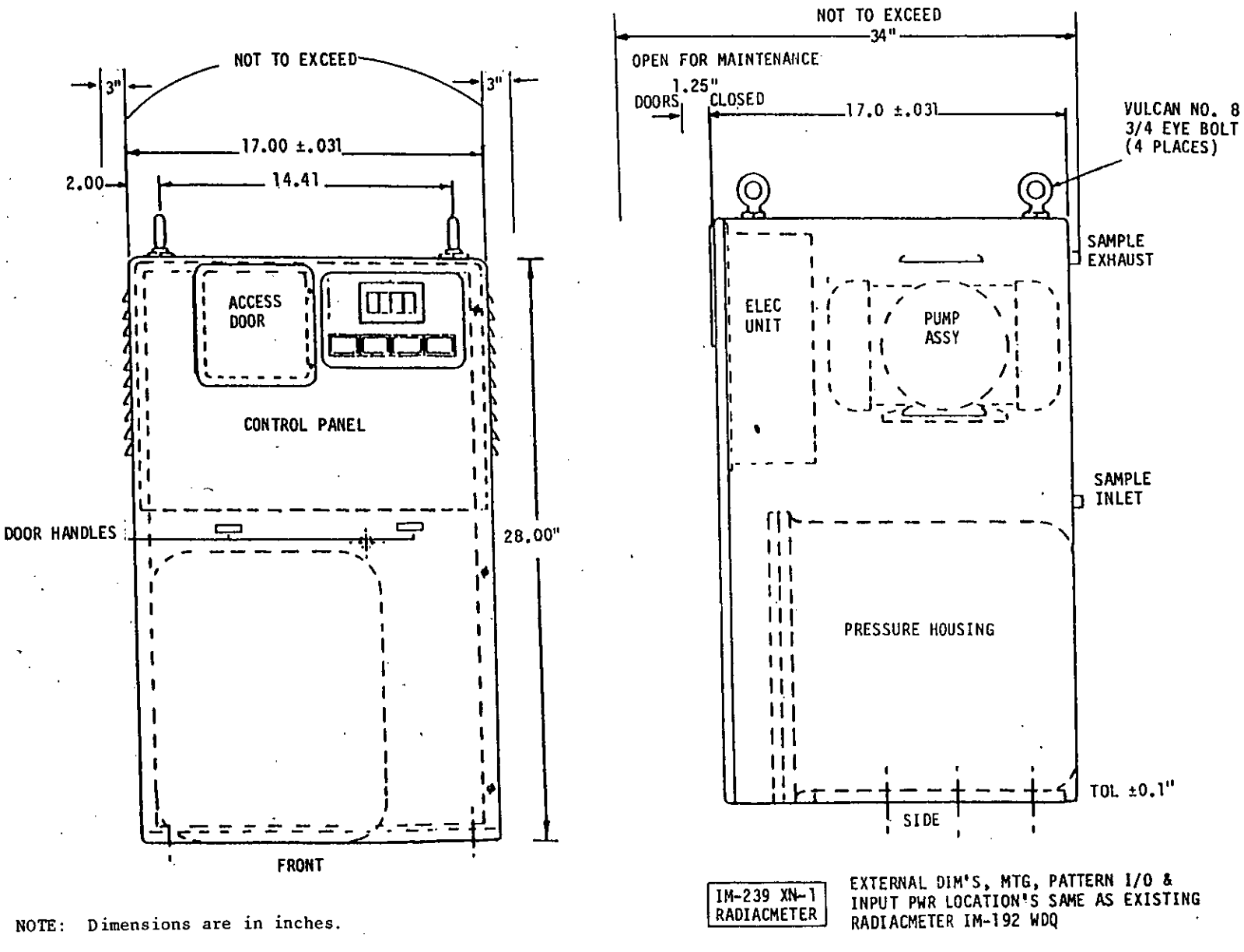
3.2.1 Radiacmeter definition. The requirements specified in 3.2.1.1 through 3.2.2.2.4 identify the radiacmeter and its relationship to the environment.

3.2.1.1 Radiacmeter diagrams. The radiacmeter outline dimensions shall be as specified in FIGURE 1. The mounting dimensions shall be as specified in FIGURE 2.

3.2.2 Interface definitions. Interface definitions shall be as specified in 3.2.2.1 through 3.2.2.2.4.

3.2.2.1 Internal interfaces. The significant internal interfaces shall be designed to minimize electrical interference and to provide proper load and source matching. Mechanical interfaces shall be designed to optimize the performance of the subsystems. Size constraints shall be considered in laying out and arranging the mechanical and electrical design.

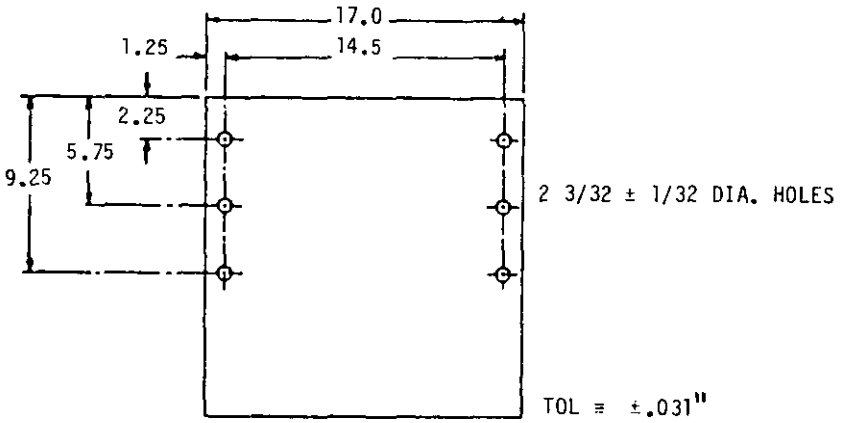
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NOTE: Dimensions are in inches.

FIGURE 1. Radiacmeter IM-239 XN-1/WDQ external dimensions and maximum envelope.

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NOTE: Dimensions are in inches.

FIGURE 2. Radiacmeter IM-239 location and size of mounting holes.

3.2.2.2 External interface. External interface shall be as specified in 3.2.2.2.1 through 3.2.2.2.4.

3.2.2.2.1 Power. Power shall be 115 volts (V), 60 hertz (Hz) ±10 percent, Type I in accordance with DoD-STD-1399, Section 300, 10 amperes (A) maximum running current.

3.2.2.2.2 Remote alarm. The remote alarm relay connections and power requirements shall be as specified in a through c:

a. Relay normally closed (NC) 115 V alternating current (VAC): 5 A alarm contact
b. Relay normally opened (NO) 115 VAC: 5 A alarm contact
c. 115 VAC: 0.4 A remote alarm power

A jack and a connector shall be provided subject to procuring agency approval for external interface and the jack shall be located near the power connector.

3.2.2.2.3 Air inlet and outlet ports. The location of air inlet and outlet ports shall be as specified in FIGURE 3. Air inlet and outlet ports shall be clearly marked.

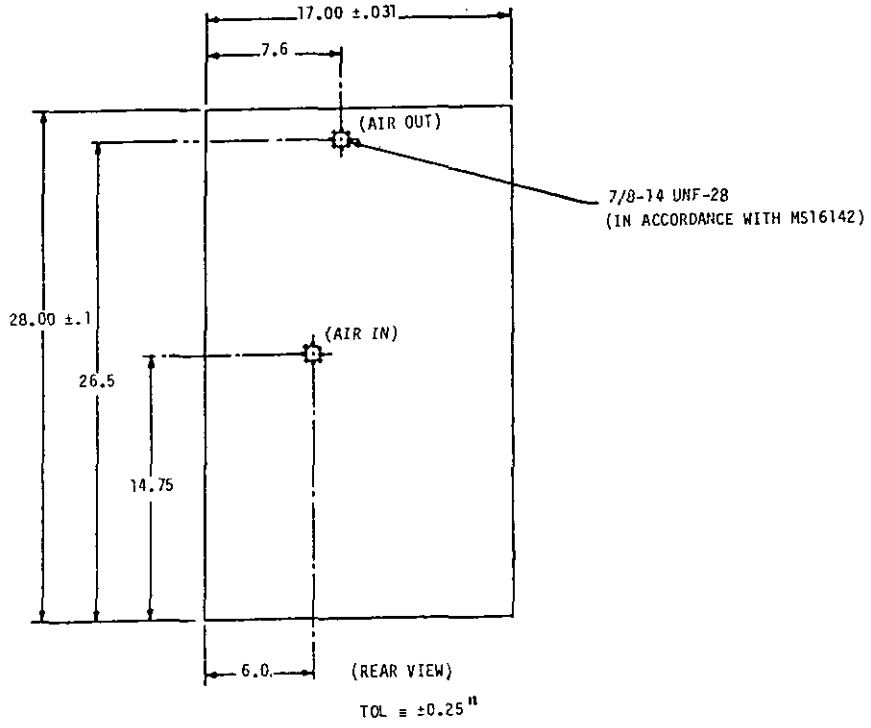
3.2.2.2.4 Space requirements. Space requirements shall be as specified in FIGURE 1.

3.3 Characteristics. Characteristics shall be as specified in 3.3.1 through 3.3.1.12.

3.3.1 Performance. The radiacmeter shall be designed for optimum response to airborne particulate emitting beta radiation. Radiacmeter detector sensitivity to alpha particles shall be retained wherever possible. All trade-off decisions between alpha and beta sensitivity shall receive concurrence from the procuring activity before incorporation in the design.

3.3.1.1 Range of response. The radiacmeter shall respond to beta particles over a range of energies. Response to cobalt (Co) 60 shall be at least 60 percent of the response to strontium (Sr) 90 and yttrium (Y) 90. Response to other isotopes having end points beta between 0.3 megaelectronvolts (MeV) and 5.0 MeV shall be between 60 percent and 150 percent of the response to Sr 90 and Y 90.

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NOTE: Dimensions are in inches.

FIGURE 3. Radiacmeter IM-239 XN-1/WGS location of air inlet and outlet ports.

3.3.1.2 Sensitivity. The radiacmeter shall have a sensitivity to Co 60 source of 8×10^{-11} micro-Curies per milliliter (μCi/ml) at the 95 percent confidence level. The measurement shall be made in a sampling time of less than two hours.

3.3.1.3 Accuracy. The radiacmeter shall have an overall accuracy of ±20 percent at and above actual measured radiation levels of 3×10^{-9} μCi/ml with a 95 percent confidence level.

3.3.1.4 Precision. The reproducibility of the system for any given measurement shall be within ±10 percent. This requirement holds for a period of at least one week after calibration at the 95 percent confidence level for actual sample beta radiation at and above 8×10^{-9} μCi/ml.

3.3.1.5 Alarm. The audible alarm shall produce a sound of 85 decibels (dB) to 90 dB at 1 meter (m) from the unit. A visual alarm indication shall also be used. The alarm shall activate when the signal exceeds the alarm threshold. The operator shall be capable of adjusting the alarm threshold at any range point from 3×10^{-10} to 8×10^{-7} μCi/ml at the front console without any special tools (see 6.4) in unit steps such as 3×10^{-10} to 4×10^{-10} and so on up to 8×10^{-7}. The alarm sounding mechanism shall be selected from MIL-A-15303.

3.3.1.6 Operating range. The range of detection shall be linear such that the total system error shall not exceed ±10 percent from actual radioactivity for each decade of range. The operating range shall run from 1×10^{-11} to 8×10^{-7} μCi/ml, from 1×10^{-11} to 1×10^{-7} μCi/ml, and ±15 percent in the last range from 1×10^{-7} to 1×10^{-6} μCi/ml. Any reading within this range shall be capable of being read on the front console.

3.3.1.7 Response time. The system shall reach 75 percent of final response to a 3×10^{-10} μCi/ml of Co 60 beta activity within 60 minutes.

3.3.1.8 Operating background. The radiacmeter shall operate in a uniform gamma background of 30 milliroentgens per hour. The energy of this gamma radiation shall be equal to or greater than that produced by Co 60. A background of 30 milliroentgens per hour shall result in no more than one false alarm in 5000 hours with the alarm set at 3×10^{-10} μCi/ml and the actual activity of less than 8×10^{-11} μCi/ml. A minimum alarm response to 40 milliroentgens per hour is desirable.

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3.3.1.9 Calibration. Calibration shall be built into the radiacmeter. The operating range point for calibration shall be selected to be near the midrange of the radiacmeter. An actual radioactive calibration source, which is automatically introduced in front of the detector, shall be used. The calibration source shall be a pure beta emitter with a stable daughter product such a technetium (Tc) 99 which is nearly equal to Co in beta end point energy and shall have an activity less than one-tenth the amount that would require Nuclear Regulatory Commission (NRC) license. The operation of the pump may be stopped or bypassed during exposure of the calibration source.

3.3.1.10. Saturation. The radiacmeter shall not indicate a response of less than 8×10^{-7} µCi/ml for any actual radioactive concentration above this level.

3.3.1.11 Particle collection efficiency. The radiacmeter shall use filter paper with an efficiency of at least 98 percent for particles with diameters greater than 0.3 micrometer.

3.3.1.12 Warm-up time. The time for the radiacmeter to stabilize within the specified requirements shall be less than 30 minutes.

3.3.2 Physical. Physical characteristics shall be as specified in 3.3.2.1 through 3.3.2.3.

3.3.2.1 Weight. The total weight of the radiacmeter shall not exceed 227 kilograms (kg) (500 pounds (lbs)).

3.3.2.2 Size. The radiacmeter shall conform to the height and depth dimensions provided in FIGURE 1.

3.3.2.3 Critical dimensions. The radiacmeter will be used as a direct replacement for the IM-192/WDQ radiacmeter as specified in MIL-R-28760. This means that certain dimensions shall be considered critical in their effect on the final design. The radiacmeter shall be designed to fit the existing mounting bolts as dimensioned in FIGURE 2. Also, air inlet and outlet ports shall use the fittings specified and located as shown in FIGURE 3.

3.3.3 Reliability. The upper test level mean-time-between-failure (MTBF) (θ_0 as defined by MIL-STD-781) shall be 5000 hours. The lower test level (θ_1 as defined by MIL-STD-781) shall be 2500 hours.

3.3.4 Maintainability. Maintainablity shall be as specified in 3.3.4.1 through 3.3.4.2.4.

3.3.4.1 Quantitative corrective maintenance. The radiacmeter, including built-in-test (BIT), shall have a mean-corrective-maintenance-time (M_{ct}) not exceeding 0.5 hour and a maximum-corrective-maintenence-time (M_{maxct}) (95th percentile) not exceeding 1.0 hour when corrective maintenance is accomplished at the organizational level of maintenance by the replacement of lowest subassemblies (modules, printed circuit boards (PCBs)) and chassis-mounted parts (electronic, electrical, electromechanical, and mechanical parts). The corrective maintenance time includes localization, isolation, disassembly, interchange, reassembly, alignment, and checkout for all corrective maintenance actions. BIT circuits are included in the corrective maintenance time requirement. The radiacmeter is considered to be in a corrective maintenance mode whenever a failure or malfunction occurs, including a malfunction that allows degraded operation.

3.3.4.2 Test measurement and diagnostic equipment (TMDE). TMDE shall be as specified in 3.3.4.2.1 through 3.3.4.2.4.

3.3.4.2.1 Test provisions. The equipment shall contain the test provisions specified in a through c:

a. Class A test provisions. Class A test provisions shall provide a means to verify the on-line equipment is operating properly. Class A provisions shall be accomplished by the use of BIT, built-in-test-equipment (BITE), or both, and shall bave the capability of detecting at least 96 percent of the equipment failures.

b. Class B test provisions. Class B test provisions shall provide a means to fault isolate 94 percent of the equipment failures to the subassembly as specified in 1 and 2:

1. 84.6 percent of the equipment failures shall be isolated to a single subassembly.
2. 9.4 percent of the equipment failures shall be isolated to two or three subassemblies.

Class B test provisions shall be accomplished when the equipment is off-line, on-line, or both, through the use of BIT, BITE, or both.

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c. Class C test provisions. Class C test provisions shall provide a means to fault isolate to a function (that is, preamplifier and discriminator, air regulation and cut-off assembly, and so forth) and to align the system or unit or assembly or subassembly.

Class C provisions may be accomplished by the use of test points in conjunction with automatic test equipment (ATE), general purpose electronic test equipment (GPETE), or special purpose electronic test equipment (SPETE).

3.3.4.2.2 BIT and BITE criteria. BIT and BITE criteria shall be as specified in a through c:

a. Protection from BIT and BITE failures. The ability of the prime equipment to perform its intended function shall not be impaired by the malfunction of BIT and BITE.

b. BIT and BITE indicators. BIT and BITE indicators shall be the equipment numeric display and MAINTENANCE WARN light. These indicators shall provide a GO or NO-GO indication at the level (subassembly, system) to which it is intended and shall be made visible by removal of no more than one cover plate, if it is covered. BIT and BITE indicators shall be located as specified in 1 and 2:

1. Class A test indicators shall be located on the control panel of the system.

2. Class B test indicators shall be located on the subassembly or system front panel.

c. BIT and BITE activation. BIT and BITE activation shall be as specified in 1 and 2:

1. BIT and BITE used in Class A test provisions shall be energized automatically without operator's initiation.

2. BIT and BITE used in Class B test provisions may be energized automatically or manually.

3.3.4.2.3 Test points criteria. Test points shall conform to the requirements specified in a through d:

a. Signal quality and circuit performance. Test points, test jacks, or both, shall be provided to permit the injection of signals and the monitoring of signals at the input/output (I/O) terminals of the subassembly.

b. Fault isolation. Test points, test jacks, or both, shall be provided to permit the injection or measurement of signals at I/O terminals of the subassembly for the purpose of isolating all faults to electrical functions (that is, integrated circuit (IC), filter, mixer, voltage controlled oscillator (VCO), and so forth). Test points and test jacks shall provide a means for performing fault isolation on the subassembly with the subassembly removed from the prime equipment.

c. Alignment. Test points, test jacks, or both, shall be provided to permit alignment of the equipment and equipment assemblies and subassemblies.

d. Calibration. Class A, B, and C test provisions shall be provided with test points, test jacks, or both, to permit calibration of the BIT and BITE functions of the equipment. Calibration of the Class A, B, and C BIT and BITE functions shall not exceed two hours.

3.3.4.2.4 GPETE. GPETE to be used at all levels of maintenance shall be selected from the standard and substitute standard equipments listed in MIL-STD-1364.

3.3.5 Environmental conditions. Environmental conditions shall be as specified in 3.3.5.1 through 3.3.5.3.4.

3.3.5.1 MIL-E-16400 application. The paragraphs of MIL-E-16400 specified in a through g shall apply:

a. Environmental service conditions (for sheltered controlled environment)
b. Nonoperating temperature range (-62°Celsius(C) to 71°C) (-80°F to 160°F)
c. Humidity
d. Shock (hard-mounted)
e. Vibration
f. Inclination (60 degrees submarine)
g. Magnetic field environment

3.3.5.2 Operating temperature range. The radiacmeter shall be capable of operating with less than five percent change in calibration and response over a temperature range of 0°C to 60°C for duration of the temperature cycling test of 4.5.16.2.

3.3.5.3 Pressure. Pressure shall be as specified in 3.3.5.3.1 through 3.3.5.3.4.

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3.3.5.3.1 Operation. The radiacmeter shall operate in 6.67×10^4 pascals to 10.67×10^4 pascals (500 torr to 800 torr, 9.67 pounds per square inch absolute (psia) to 15.47 psia) with a maximum differential pressure between the sealed sampling system (see 3.3.5.3.2) and surrounding air of 1 inch of mercury (0.5 psia).

3.3.5.3.2 Sealed systems. The radiacmeter shall have a sealed sampling system capable of processing and returning sampled air from an area without significant leakage to the surrounding area as specified herein. The sealed sampling system shall be designed to withstand internal air pressure up to 70 pounds per square inch gauge (psig) at 24°C (75° Fahrenheit (F)), with leakage from the sealed sampling system to surrounding air not in excess of 15 standard liters per hour. The sealed sampling system shall also be designed to contain the following steam pressurization transient. The sealed sampling system of a radiacmeter operating in an ambient temperature of not less than 38°C (100°F) shall contain pressurized steam at 22 psig, 113°C (235°F).

3.3.5.3.3 Pressure relief. A pressure relief valve shall be used to relieve pressures within the sealed sampling system greater than 70 psig. The relief valve shall actuate at a pressure of 70-0+10 psig and shall activate and latch an audible alarm when in operation. A pressure vent valve, clearly labeled and accessible within the radiacmeter cabinet, shall be provided to relieve pressure in the sealed sampling system during maintenance and prior to opening the pressure housing door.

3.3.5.3.4 Steam transient. The sealed sampling system shall be designed to contain the following steam pressurization transient: The sealed sampling system of a radiacmeter operating in an ambient temperature of not less than 38°C (100°F) shall contain pressurization steam of 22 psig at 113°C (235°F).

3.3.6 Transportability. The radiacmeter shall be designed to withstand the environmental problems resulting from normal transportation and handling without damage or calibration change. Calibration and alignment changes that can be corrected following normal installation procedures by using only the front operator's controls, will be acceptable.

3.4 Design and construction. Design and construction shall be as specified in 3.4 through 3.4.9.9.

3.4.1 Parts, materials, and processes. Parts, materials and processes shall be as specified in 3.4.1 through 3.4.1.5.

3.4.1.1 Applicable MIL-E-16400 paragraphs. All paragraphs of the Parts, materials, and processes paragraph of MIL-E-16400 shall apply except as specified in a and b:

a. Tube shields
b. Transmission lines and related radio frequency (RF) parts

3.4.1.2 Capacitors. Capacitors shall be derated at least 50 percent; for example, the actual maximum peak voltage that the component experiences shall be not greater than 0.5 times the specification voltage.

3.4.1.3 Resistors. Resistors shall be derated at least 50 percent; for example, the actual wattage of the component in the circuit shall be not greater than 0.5 times the specification wattage rating as calculated for the ambient condition.

3.4.1.4 Stress derating. The contractor shall establish and maintain specific criteria and procedures acceptable to the Government for stress derating of electrical and mechanical parts. Criteria shall be stated in terms of applied versus rated voltage, power, junction temperature, strain, and so forth, as applicable for each part type. Procedures shall be implemented which will ensure that the established derating criteria are adhered to during the design process.

3.4.1.5 Parts control. The parts to be incorporated in the equipment shall be controlled in accordance with MIL-STD-965, Procedure I.

3.4.2 Electromagnetic compatibility (EMC). The radiacmeter shall conform to the requirements of MIL-STD-461, Class A4, above-deck test equipment, with the exceptions specified in a and b:

a. CS06. In addition to the 400-V test, a 100-V peak with 0.5 millisecond width shall be applied at a repetition rate of 360 pulses per second.
b. RS03. Upper test frequency shall be 40 gigahertz (GHz).

MIL-HDBK-241 may be used for guidance in EMI reduction in power supply design.

3.4.3 Thermal design. The thermal design shall be in accordance with MIL-E-16400.

3.4.4 Identification and marking. Identification and marking shall conform to the Identification and marking paragraph of MIL-E-16400.

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3.4.5 Workmanship. Workmanship shall be as specified in 3.4.5.1 and 3.4.5.2.

3.4.5.1 General workmanship. Workmanship and soldering shall conform to the requirements of MIL-E-16400.

3.4.5.2 Workmanship screen. All production radiacmeters produced in accordance with this specification shall withstand a defect detection vibration screen of random type vibration at $0.04g^2/Hz$ ±3 dB from 80 Hz to 350 Hz and temperature cycling as specified in 4.5.16.2.2.

3.4.6 Interchangeability. Major assemblies specified in 3.4.9 shall be interchangeable. Adjustment and calibration shall be allowed after an interchange has been made.

3.4.7 Safety. Safety requirements shall be in accordance with the Safety criteria paragraph of MIL-E-16400, except as specified in 3.4.7.1 through 3.4.7.10.

3.4.7.1 Radiacmeter power connections. Radiacmeter power connectors for 115 VAC input power and interconnecting alternating current power cable connectors between radiacmeters shall have only three pins designated A, B, and C. Pin assignment and conductor color code within the radiacmeter shall be as follows:

Pin designation	Conductor assignment	Color code
A	115 VAC return	White or light blue
B	Safety ground	Green or green with yellow stripe
C	115 VAC high (hot)	Black or brown

Black or brown and white or light blue color code shall be maintained from the input connector pins to all components having the same voltage and frequency as the input power.

3.4.7.2 Circuit breakers and fuses. Circuit breakers and fuses and fuseholders shall conform to the Circuit breakers and Fuses and fuseholders paragraphs of MIL-E-16400.

3.4.7.3 Overcurrent device location. A fuse or an overcurrent trip unit of a circuit breaker shall be connected in series with each ungrounded conducter as determined by the National Electrical Code (see NFPA 70-1983) for grounded source power and shall be connected on the load side of the ON-OFF power switch. The ON-OFF switch may be the circuit breaker.

3.4.7.4 Material flammability. Material used as part of the radiacmeter shall conform to the requirements of MIL-E-16400.

3.4.7.5 Radiacmeter electrical power ON-OFF switch. A switch for disconnecting radiacmeters from all electrical power systems shall be mounted on the radiacmeter front panel and the function clearly labeled. The switch shall break all power conductors.

3.4.7.6 Power indicator lamp. A green power indicator lamp shall be mounted near the switch to indicate when the radiacmeter is energized. The lamp shall be connected to the load side of the switch and across the input power conductors. The lamp may be illuminated directly from the 28 V power supply.

3.4.7.7 Power supply protection. Fault conditions ranging from open circuits to short circuits shall cause no damage to the power supply.

3.4.7.8 Printed circuit asembly protection. With power applied to the radiacmeter, removal or insertion of printed circuit assemblies shall not damage or degrade any part of the radiacmeter. Each assembly shall be keyed to prevent insertion into the wrong connector and to prevent reversal of insertion.

3.4.7.9 Human error design protection. The design of the radiacmeter shall prevent damage as a result of operator errors involving procedures in control sequencing, adjustments, and in responding to alarms and fault indicators.

3.4.7.10 Mechanical hazards. All accessible corners shall be rounded. Exterior doors, hinged covers, and drawers shall have automatic latch and quick release devices which shall be actuated before they can be opened and stops to hold them in the open position for each maintenance configuration. Radiacmeter design shall include provisions to prevent accidental derailing, opening and pulling out of drawers of rack-mounted equipment. Design shall include means to prevent accidental dropping of exterior covers and panels during removal and attachment. The small control access door may be designed without bracket to hold the door in open position.

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3.4.8 Human engineering. Human engineering shall comply with the human engineering operational and maintenance design criteria of MIL-STD-1472 and the requirements specified in 3.4.8.1 and 3.4.8.2.

3.4.8.1 Controls, indicators, and panel layouts. The design of controls and indicators and their arrangement on operator and maintainer panels shall conform to the criteria of MIL-STD-1472 and shall be in accordance with the Controls, indicators and panel layout paragraph of MIL-E-16400. The color coding and marking of pin jacks and other test points shall be in accordance with MIL-STD-415. Abbreviations used on panels shall be in accordance with MIL-STD-12. All controls shall be readily operated by a standing operator with the unit mounted on the floor or on a pedestal which is between 0 m and 1 m high.

3.4.8.2 Display. The display shall provide a visual readout of all required display functions that include radiation level, alarm indication and level, flow rate, high or low flow alarm, malfuction indication and diagnostics, calibration level and indications, and low paper warning. The indicators shall be readable at standing eye level, up to a distance of 3 m with the unit located anywhere between the floor and 1 m off the floor.

3.4.9 Major assemblies. The major assemblies shall be as specified in a through o:

a. Paper feed assembly
b. Detector assembly
c. High voltage (HV) power supply and preamplifier assembly
d. Input power and control assembly
e. Latch standard electronic modules (SEM)
f. Divider SEM
g. Resistor SEM
h. Multiplex SEM
i. Buffer SEM
j. Counter SEM
k. Science Applications Inc. (SAI) microprocessor SEM, or equivalent
l. Low voltage (LV) power supply assembly
m. Pump motor assembly
n. Flow meter assembly
o. Pressure housing and shield assembly

3.4.9.1 Paper feed assembly. The paper feed assembly shall move the filter paper at a constant speed as the sampled air is filtered through. The paper feed assembly shall have a fast advance capability to clear the tape area and to allow for testing. The paper feed assembly shall also have a stationary positon for testing and measuring paticulate samples. The paper drive shall have an indicator visible on the operator's panel that shall indicate when less than 24 hours of filter tape remains. A minimum 30-day supply of filter tape shall be provided.

3.4.9.2 Detector assembly. The detector assembly shall be of minimum size yet still provide the sensitivity required herein.

3.4.9.3 HV power supply and preamplifier assembly. The HV power supply shall provide filtered and regulated power to the detector.

3.4.9.4 Input power and control assembly. The controls shall be included on the front of the unit to perform the necessary functions specified herein. The controls include alarm setting and resetting, calibration, air flow, paper speed, ON-OFF power, required test modes, malfunction diagnostics, lamp check for display, pump power disengage, and gain setting.

3.4.9.5 SEM assemblies. SEM assemblies as specified in MIL-E-16400 shall be used to the maximum extent practical in the design.

3.4.9.6 SAI microprocessor SEM. The SAI microprocessor shall be a special SEM conforming to the requirements of MIL-E-16400 and shall contain the logic assembly. The logic assembly shall be a microprocessor-based system that receives electronic signals from the amplifier and discriminator and processes the electronic signals to determine the radiation level. An alarm function shall be activated when a preset radiation level is exceeded. The signal processor shall also keep a continuous check on all operating electronic circuitry and shall warn when a circuit is malfunctioning and may be used to aid in troubleshooting the faulty electronics.

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3.4.9.7 LV power supply assembly. The LV power supply shall provide filtered and regulated power to the electronics assemblies including the amplifiers, signal processing, and display, but excluding the pump motor. The LV power supply shall be designed to conform to all line noise and other interference problems experienced in a shipboard environment as specified herein. Power line noise as specified herein shall not interfere with the electronic processing of the radiacmeter.

3.4.9.8 Pump motor assembly. The pump motor assembly shall be one replaceable unit that can be removed and replaced in one-half hour. The pump selected shall have the minimum weight, size, and drive power requirements and yet have adequate and volume moving capabilities to conform to the minimum detection requirements specified herein.

3.4.9.9 Flow meter assembly. The flow of the air shall be measured within an accuracy of ±5 percent. The flow meter assembly shall provide air flow measurement data to the microprocessor, which shall be used to correct the radiation measurement displayed to the operator.

3.5 Enclosure. Degree of enclosure shall be dripproof (45 degrees) in accordance with MIL-STD-108.

3.6 Airborne and structureborne noise. The airborne and structureborne noise shall conform to the requirements of MIL-E-16400.

3.7 Finish. Surfaces shall be in accordance with MIL-E-16400.

3.8 Computer resources. Computer resources shall be as specified in 3.8.1 through 3.8.3.

3.8.1 Resource allocation and reserves. The Resource allocation and reserve requirements paragraph of DoD-STD-1679 shall be applied to computer software and firmware development. The 20 percent reserves, as specified in the Resource allocation and reserve paragraph of DoD-STD-1679, are exclusive of growth requirements.

3.8.2 Software generation. The software shall conform to the requirements specified in the Software generation paragraph of DoD-STD-1679.

3.8.3 Computer programming standards and conventions. The radiacmeter software product shall conform to the Programming standards and Programming conventions paragraphs of DoD-STD-1679.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of, but is not limited to, a) surveillance of the operations to determine that practices, methods, and procedures of the written quality program are being properly applied, b) Government product inspection to measure quality of the product to be offered for acceptance, and c) Government inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been taken or until conformance of the product to prescribed criteria has been demonstrated.

4.1.2 Quality assurance terms and definitions. Quality assurances terms used in this specification shall be as defined in MIL-STD-109.

4.2 Classification of inspections. The inspection requirements specified herein are classified as specified in a through c:

a. First article inspection (see 4.3)
b. Quality conformance inspection:
   1. Production inspection (see 4.4.1)
c. Inspection of preparation for delivery (see 4.8)

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4.3 First article inspection. Unless otherwise specified (see 6.2), four units shall be required for first article inspection. First article inspection shall consist of all examinations and tests necessary to determine compliance with the requirements of this specification. First article inspection shall include the tests specified in TABLE I.

TABLE I. First article inspection.

Examination or test	Requirement paragraph	Test paragraph
Surface examination		
Weight	3.3.2.1	4.5.1
Size	3.3.2.2, 3.3.2.3	4.5.1
Parts and materials	3.4.1	4.5.1
Finish	3.7	4.5.1
Marking	3.4.4	4.5.1
Safety	3.4.7	4.5.1
General workmanship	3.4.5.1	4.5.16.1
Workmanship screen	3.4.5.2	4.5.16.2
Preperformance	3.1	4.5.1
Performance		
Range	3.3.1.1	4.5.2
Sensitivity	3.3.1.2	4.5.3
Accuracy	3.3.1.3	4.5.4
Precision	3.3.1.4	4.5.5
Alarm	3.3.1.5	4.5.6
Operating range	3.3.1.6	4.5.7
Response time	3.3.1.7	4.5.8
Operating background	3.3.1.8	4.5.9
Calibration	3.3.1.9	4.5.10
Saturation	3.3.1.10	4.5.11
Warm-up time	3.3.1.12	4.5.12
Pressure	3.3.5.3	4.5.17
Temperature		
Low temperature	3.3.5.2	4.5.1
High temperature	3.3.5.2	4.5.1
Humidity	3.3.5.1	4.5.13
Shock 1/	3.3.5.1	4.5.1
Vibration	3.3.5.1	4.5.1
Inclination	3.3.5.1	4.5.1
Airborne and structureborne noise	3.6	4.5.1
Enclosure	3.5	4.5.1
Weld	3.1	4.5.1
Steady-state voltage and frequency	3.2.2.2.1	4.5.1
Transient voltage	3.2.2.2.1	4.5.1
Spike voltage	3.2.2.2.1	4.5.1
Power interruption	3.2.2.2.1	4.5.1
Power and power factor	3.2.2.2.1	4.5.1
Insulation resistance	3.2.2.2.1	4.5.1
Corona measurement	3.2.2.2.1	4.5.1
Leakage current	3.1	4.5.15
Dielectric withstanding voltage	3.1	4.5.1
Thermal design	3.4.3	4.5.1
Software acceptance	3.8	4.7
Interchangeability	3.4.6	4.5.18
EMC	3.4.2	4.5.14

1/ One unit only may be subjected to the shock test.

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4.4 Quality conformance inspection. Quality conformance inspection shall be as specified in 4.4.1.

4.4.1 Production inspection. Production inspection shall be made on every unit offered for delivery. Production inspection shall comprise such examinations and tests which will prove the workmanship and reveal omissions and errors of the production process such as, functional and performance tests at a limited number of points in the required range, tests which detect deviations from design, tests of controls and adjustments, and tests which detect hidden defects of materials. Production inspection shall include the examinations and tests specified in TABLE II.

TABLE II. Production inspection.

Examination or test	Requirement paragraph	Test paragraph	Sample size
Surface examination			
Weight	3.3.2.1	4.5.1	All
Size	3.3.2.2, 3.3.2.3	4.5.1	All
Parts and materials	3.4.1	4.5.1	All
Finish	3.7	4.5.1	All
Marking	3.4.4	4.5.1	All
Safety	3.4.7	4.5.1	All
General workmanship	3.4.5.1	4.5.16.1	All
Workmanship screen	3.4.5.2	4.5.16.2	All
Preperformance	3.1	4.5.1	All
Performance			
Range	3.3.1.1	4.5.2	All
Sensitivity	3.3.1.2	4.5.3	All
Accuracy	3.3.1.3	4.5.4	All
Alarm	3.3.1.5	4.5.6	2 per lot
Operating range	3.3.1.6	4.5.7	2 per lot
Operating background	3.3.1.8	4.5.9	2 per lot
Calibration	3.3.1.9	4.5.10	All
Pressure	3.3.5.3	4.5.17	All
Airborne and structure-borne noise	3.6	4.5.1	2
Weld	3.1	4.5.1	All
Steady-state voltage and frequency	3.2.2.2.1	4.5.1	2
Power interruption	3.2.2.2.1	4.5.1	2
Power and power factor	3.2.2.2.1	4.5.1	2
Leakage current	3.1	4.5.15	2 per lot
EMI	3.4.2	4.5.14	2 per lot

4.5 Test methods. Test methods shall be as specified in 4.5.1 through 4.5.18.1.

4.5.1 Examinations and tests. Examinations and tests specified in TABLES I and II shall be conducted in accordance with the applicable test methods of MIL-E-16400, except as specified in 4.5.2 through 4.5.18.1.

4.5.2 Range. Radioactive point sources of Co 60, Sr-90 and Y-90, thallium (Tl)-204 and ruthenium (Ru)-106 having an activity that will read between 1 x 10^{-9} and 1 x 10^{-7} micro-Curies per cubic centimeter (μCi/cu cm) on the radiacmeter display shall be used. The actual activity of these sources shall be traceable to National Bureau of Standards (NBS) standards with a calibration accuracy of ±5 percent. The source may be deposited in a test source holder; however, the test conditions shall reproduce actual measurement conditions including back-scattering structural metals used, and distance losses. The actual readings of the radiacmeter shall be compared to the actual activity of the sources to determine the efficiencies of each source. The efficiency of each source shall then be compared to Sr-90 and Y-90 by dividing the efficiency of each by the efficiency of Sr-90 and Y-90. The result shall be greater than 60 percent and less then 150 percent of the Sr-90 and Y-90 efficiency.

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4.5.3 Sensitivity. A test source of Co 60 shall be used with activity equivalent to the maximum equilibrium amount deposited on the filter paper as if the radiacmeter were in normal operating mode, sampling air containing a static radioactivity of 8×10^{-11} µCi/ml. A test source of less activity may be used and the source may be setup in a source holder provided that the holder accurately reproduces the actual sample measurement conditions, including the effects of backscattering and distance from the detector. To determine the actual reading, the background shall be determined and subtracted from the reading provided by the radiacmeter when the test source is in place. The actual reading shall be within 25 percent of the reading expected from the test source, which is equal to or less than 8×10^{-11} µCi/cu cm. The calculation of this activity shall assume that the radiacmeter is operating in a uniform concentration of 8×10^{-11} µCi/ml for a period of 2 hours.

4.5.4 Accuracy. A radioactive source similar to the source of 4.5.3, except equivalent to a static radioactivity of $4 \pm 1 \times 10^{-10}$ µCi/ml shall be used. The radiacmeter shall be calibrated using normal operational procedures, then run for at least 4 hours before the source is placed in front of the detector. The reading of the radiacmeter shall be within ±20 percent of the true value of the source at the 95 percent confidence level. The source shall be traceable to NBS within ±5 percent.

4.5.5 Precision. A radioactive source as specified in 4.5.3, except with known activity of $9 \pm 1 \times 10^{-9}$ µCi/ml, shall be measured by a calibrated equipment. The radiacmeter shall be run continuously for a week. A sufficient number of readings shall be taken with no calibration after the start of this time interval to demonstrate a precision of ±10 percent ($\pm 1 \times 10^{-9}$ µCi/ml) with a 95 percent confidence level. Other tests may be performed during this period, subject to approval of the procuring activity.

4.5.6 Alarm. Each of the test sources used for the operating range test specified in 4.5.7 shall be placed in front of the detector with the alarm set 10 percent lower than the actual display reading of each source. The alarm shall activate for each setting. The sound level at 1 m in dB of the alarm shall be measured and shall be between 85 dB and 90 dB.

4.5.7 Operating range. Three Co 60 test sources which will produce display readings on the radiacmeter of $5 \pm 3 \times 10^{-10}$, $5 \pm 3 \times 10^{-9}$, and $5 \pm 3 \times 10^{-8}$ µCi/cu cm shall be used. The sources may be setup in source holders that accurately reproduce the sampling conditions including the effects of backscattering and distance from the detector. When inserted in front of the detector, each source shall produce a display reading that is within ±10 percent of the actual reading expected. Background effects shall be subtracted. The source material shall be NBS traceable within a ±5 percent calibration error at the 95 percent confidence level.

4.5.8 Response time. A calibrated distributed source of Co 60 NBS traceable to ±5 percent at the 95 percent confidence level with actual activity of $3 \pm 2 \times 10^{-10}$ µCi/ml shall be used. The source may be placed in a source holder as specified in 4.5.7. The source holder shall be moved in front of the detector slowly or in small steps of 10-minute intervals at the same rate the paper would move. The activity displayed at the end of 1 hour shall be at 75 percent of the actual activity level measured as if the system had been running continuously in a static level of activity as calculated from the true activity of the calibration source used.

4.5.9 Operating background. The radiacmeter shall be placed in a 30 milliroetgen per hour Co 60 gamma-background with the alarm set at 3×10^{-10} µCi/ml. The count rate shall be measured and statistically added to a count rate equivalent to 8×10^{-11} µCi/ml. The resultant statistical count rate shall indicate a MTBF of at least 5000 hours. The radioactive source used for the test shall be placed at the front, back, and both sides. The highest radiacmeter display reading shall be used to calculate the false alarm MTBF.

4.5.10 Calibration. The calibration source installed in the radiacmeter shall be inspected and tested to ensure that it is Tc-99 with a display activity of $3 \pm 2 \times 10^{-8}$ µCi/cu cm. The source's actual activity shall be checked to ensure that it is less that one-tenth the amount of activity that would require an NRC license. The check source used for the precision test specified in 4.5.5 shall be used for this test. The reading of the radiacmeter check source shall be compared to the precision check source. At least three sets of readings shall be taken and shall be within ±10 percent of each other.

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4.5.11 Saturation. A radioactive source shall be introduced in front of the detector that would cause the radiacmeter to register at least 8 x 10^{-6} µCi/ml. The reading shall remain at or above 8 x 10^{-7} µCi/ml and not decrease.

4.5.12 Warm-up time. The radiacmeter shall be energized for exactly 30 minutes after which the unit shall be calibrated using built-in check source. This calibration shall be checked using built-in check source for 8 hours and shall remain within ±10 percent of the original calibration. The flow rate shall remain within ±10 percent for the time period.

4.5.13 Humidity test. Humidity tests shall be conducted in accordance with Method 507.1, Procedure IV of MIL-STD-810C, except: add to Step 9, inspect for defects in accordance with the Visual examination paragraph of MIL-E-16400. Blistering of organic coating shall be cause for rejection.

4.5.14 EMC test. The requirements of 3.4.2 shall be verified by tests performed in accordance with MIL-STD-462. For production inspection only, tests CE03, CS06, and RE02 shall be performed.

4.5.15 Leakage current test. The leakage current test shall be as specified in 4.5.15.1 and 4.5.15.2

WARNING

THIS TEST MAY BE HAZARDOUS DUE TO THE UNGROUNDED CONDITION OF THE RADIACMETER DURING THE TEST. DO NOT TOUCH EXPOSED METAL SURFACES.

THE UNITED STATES GOVERNMENT NEITHER ASSUMES NOR ACCEPTS RESPONSIBILITY FOR ANY INJURY OR DAMAGE THAT MAY OCCUR DURING OR AS A RESULT OF THIS TEST.

4.5.15.1 Radiacmeter connections. Each radiacmeter directly connected to an external power source and units deriving power from the radiacmeter shall be placed on an insulated surface. All safety ground conductors between the radiacmeter and units deriving power from the radiacmeter shall be intact. The safety ground conductor between the radiacmeter under test and the source power shall be opened during the test. The radiacmeter shall be connected as shown in FIGURE 4 if it is connected to single-phase source power. When direct current (DC) measurements are performed, do not reverse polarity unless the equipment has reverse polarity protection.

4.5.15.2 Measurement. Leakage current shall be measured on the radiacmeter in its normal operating configuration. Radiacmeter controls in each operating mode shall be such that maximum power will be utilized during leakage current measurements. The leakage current shall be determined by the voltage-drop method. A true root-mean-square (rms) voltmeter shall be used. The voltage measured across the 1500-ohm resistor shall not exceed 7.5 V at the highest nominal power line voltage and the highest and lowest nominal power line frequencies for which the radiacmeter is designed. The overall measurement error shall not exceed five percent. The probe shall be used on all external conducting parts such as case, connector housings, recessed calibration or adjustment controls, and control shafts with knobs removed, and the voltage measured for every combination of switch positions available in FIGURE 4. The open safety ground conductor shall be reconnected immediately after the test is completed.

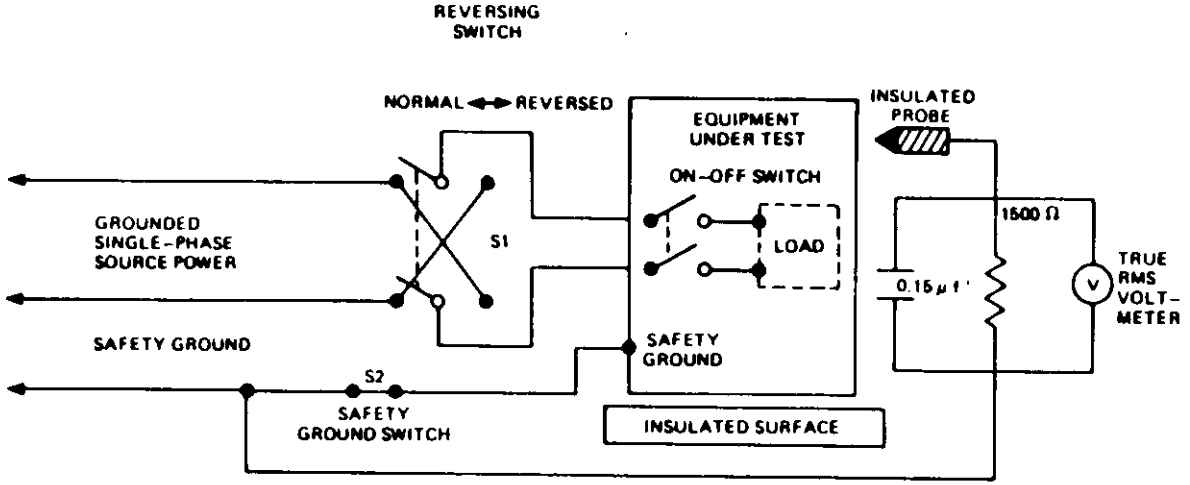
4.5.16 Workmanship. Workmanship shall be as specified in 4.5.16.1 and 4.5.16.2.2.

4.5.16.1 General workmanship. The radiacmeter, including subassemblies and assemblies, shall be examined for workmanship and soldering during the fabrication and assembly process for conformance to the requirements of 3.4.5.1.

4.5.16.2 Workmanship screen. Prior to conducting temperature cycling (see 4.5.16.2.2) vibration (see 4.5.16.2.1) shall be performed on each radiacmeter. The vibration may be performed at the module, drawer, or end item level. All the hardware, including cables and connectors, shall be exposed to vibration.

4.5.16.2.1 Random vibration. The vibration shall be random, or subject to procuring activity approval, pseudo-random or complex waveform vibration, for an accumulated time of 10 minutes in the axis deemed most susceptible to vibration excitation. All test radiacmeters shall be hard-mounted (without shock isolators) and subjected to the vibration conditions of FIGURE 5. The control accelerometer shall be located next to one of the mounting points of the radiacmeter under test. Radiacmeters having a bandwidth no greater than 10 Hz for vibration frequencies up to 500 Hz and 100 Hz for vibration above 500 Hz shall be used for the control and analysis of the acceleration spectral density (ASD). The test radiacmeter shall be energized during vibration and appropriate input signal applied to observe any abnormal conditions of the output functional characteristics. All failures occurring during test shall be corrected and the test resumed.

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GENERAL ORDER OF TEST:

1. SOURCE POWER OFF. CONNECT EQUIPMENT PER DIAGRAM.
2. ON-OFF SW OFF. S1 SW NORMAL. S2 SW CLOSED. CONNECT SOURCE POWER
3. OBSERVE WARNING. S2 SW OPEN. ON-OFF SW ON.
4. FOR EACH PROBE POINT, RECORD VOLTMETER READING (CASE, CONNECTORS, CONTROLS, SHAFTS).
5. ON-OFF SW OFF. REPEAT STEP 4.
6. S1 SW REVERSED. ON-OFF SW ON. REPEAT STEP 4.
7. ON-OFF SW OFF. REPEAT STEP 4.
8. S2 SW CLOSED. S1 SW NORMAL.
9. REPEAT STEP 3 THRU 8 FOR EACH MODE OF OPERATION.
10. REMOVE SOURCE POWER. DISCONNECT EQUIPMENT. SIGN RECORD SHEET.

WARNING
DO NOT TOUCH EXPOSED METAL SURFACES

THIS TEST MAY BE HAZARDOUS DUE TO THE UNGROUNDED CONDITION OF THE EQUIPMENT DURING THE TEST. THE UNITED STATES GOVERNMENT NEITHER ASSUMES NOR ACCEPTS RESPONSIBILITY FOR ANY INJURY OR DAMAGE THAT MAY OCCUR FROM THE USE OF THIS DIAGRAM FOR LEAKAGE CURRENT MEASUREMENT.

FIGURE 4. Single-phase test diagram for leakage current measurement.

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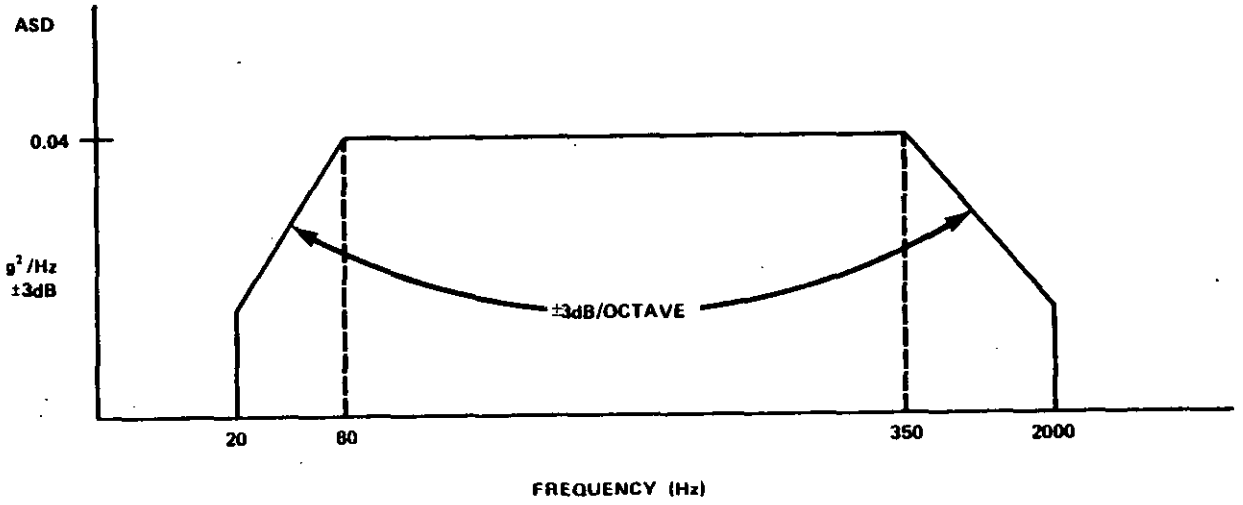


FIGURE 5. Random vibration curve

4.5.16.2.2 Temperature cycling. Each radiacmeter shall be subjected to the temperature curve of FIGURE 6. The temperature rate of change shall not be less than 5°C per minute. Radiacmeter power shall be turned on and off at the indicated times. When practical, the radiacmeter drawers, panels, and enclosures shall be opened or removed for maximum exposure to the changing temperature. When performance measurements are called for, a functional operating test shall be performed. The dwell time shall be 80 percent of the time required for the largest electrical or electronic part to become temperature stabilized. Temperature stabilization shall be as defined in MIL-STD-810C for the nonoperating condition. When failures occur, the radiacmeter shall be reworked and the cycling continued for a cumulative total of 10 cycles.

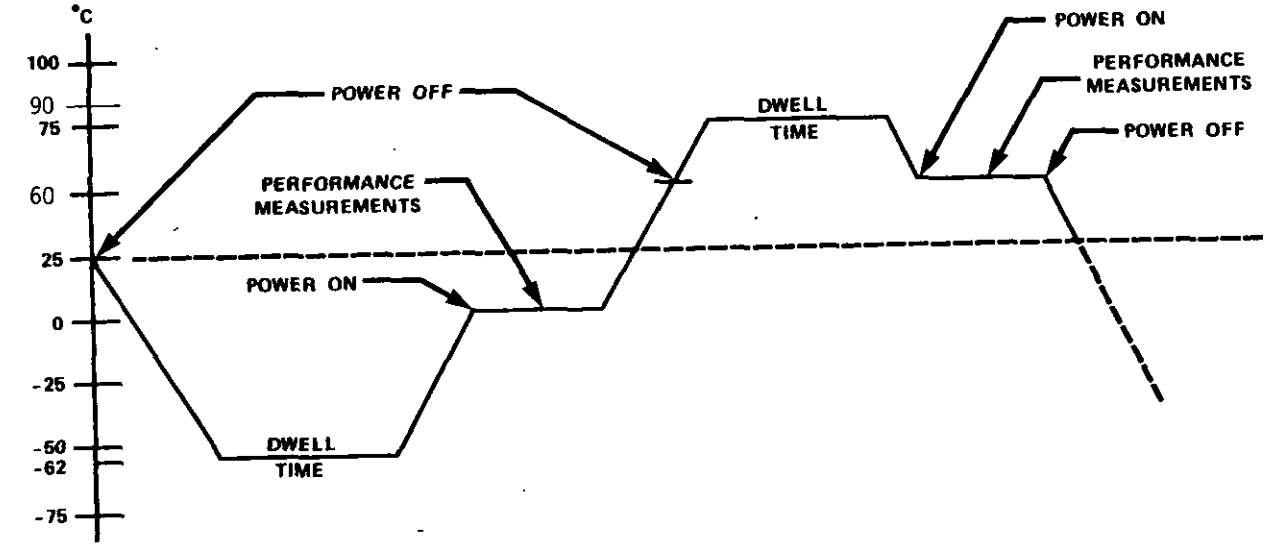


FIGURE 6. Temperature curve (one cycle)

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4.5.17 Pressure test. The radiacmeter shall be tested under various pressure conditions to determine if the equipment can hold the pressure as specified herein.

4.5.17.1 Sealed system test. The radiacmeter shall be connected to a closed pressure system containing 70 psig air pressure at 24°C (75°F) and shall not exceed a leakage rate of 15 liters per hour.

4.5.17.2 Pressure differential and altitude. The radiacmeter shall be operated between 500 torr and 800 torr and with a differential pressure of 1 inch of mercury without indication of any failures or degradation of operation.

4.5.17.3 Steam test. The sealed sampling system in surrounding air of 100°F shall be pressurized to 22 psig with 235°F steam. After reaching 22 psig, the radiacmeter shall be isolated and the pressure and temperature within the sealed sampling system allowed to decay by heat loss to surrounding air. At approximately 20 minutes after pressurization, the accessible parts of the sealed sampling system shall be inspected for evidence of leakage of steam or moisture. No detectable leakage is permitted.

4.5.18 Interchangeability test. An interchangeability test shall be performed to ensure that the radiacmeter can be installed in the same mountings as an IM-192/WDQ radiacmeter.

4.5.18.1 Interchangeability demonstration. The radiacmeter shall be examined and tested to verify and demonstrate compliance with 3.4.6 by utilizing at least two radiacmeters and interchanging each interchangeable assembly, subassembly, module, and so forth, between the two radiacmeters. Satisfactory radiacmeter performance shall be demonstrated prior to and after the interchange of the interchangeable assemblies, subassemblies, modules, and so forth. Interchangeability of piece parts shall be verified by examination.

4.6 Reliability test. When required by the procuring activity (see 6.2), the reliability test shall be conducted in accordance with the requirements of MIL-STD-781. Test Plan XXC, test condition for Category 3A, shall be used on the number of acceptance equipments specified in the contract (see 6.2) for a total of 15,000 equipment hours. No one equipment shall be run for less than 1850 hours.

4.7 Software acceptance testing. Software acceptance testing shall be conducted in accordance with the Software acceptance paragraph of DoD-STD-1679 and Government-approved test plan, test specifications, and test procedures to ensure conformance to the formal testing requirements.

4.8 Inspection of preparation for delivery. Inspection shall be conducted to ensure conformance with the requirements of Section 5.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation, packaging, packing, and marking. Unless otherwise specified herein, preparation for delivery shall be in accordance with the applicable levels of preservation, packaging, packing, and marking specified in MIL-E-17555 (see 6.2).

6. NOTES

6.1 Intended use. The radiacmeter covered by this specification is intended for use on board ships and in shore facilities when continuous monitoring of airborne particulate radioactivity is required.

6.2 Ordering data. Procurement documents should specify:

a. Title, number, and date of this specification
b. Number of first article samples to be submitted other than specified in 4.3
c. Levels of preservation, packaging, packing, and marking (see 5.1)
d. If the reliability test is to be conducted in accordance with MIL-STD-781 (see 4.6)

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6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the Defense Acquisition Regulation. The first article should be a first production item. The first article should consist of four units. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for examinations, tests, and approval of the first article.

6.4 Special tools. Special tools are defined as those tools not listed in the National Supply Catalog (copies of this catalog may be consulted in the office of the Defense Contract Administration Service).

Preparing activity:
NAVY-EC

(Project 6665-N452(EC))

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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